

Data Science Toolbox Portfolio Questions

05 Unsupervised learning - Anomalies, p-values and Missingness

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Block 5

Portfolio 05

Portfolio Question 5.1.P - Anomaly Detection Methods

- What methods for anomaly detection are there? When should you use them?
- Give an explanation of some methods that are easily implemented in practical examples.

Portfolio Question 5.2.P - Conformal prediction

- We claimed that conformal prediction could guarantee **any** model would get calibrated probabilities!
 - We only claimed it for marginal coverage. What is meant by this? When is it useful or not?
 - What methods are available for conditional coverage?
 - What data do we need to make it work?

Portfolio Question 5.3.P - Missingness

- What conclusions can we draw from this analysis about the effect of the covariates on the outcome?
 - How do we know that imputation has done a good job?
 - If the imputation went badly, how wrong could our estimates be?
- Design an experiment to try to answer these questions, by performing an evaluation of the performance of the imputation procedure. To do this, you will need to leave further data out as cross validation, or make adversarial imputation. Include your code and any non-essential figures as an appendix.